Machining Processes: Turning

Groover M.P.: Chapters 18 and 19

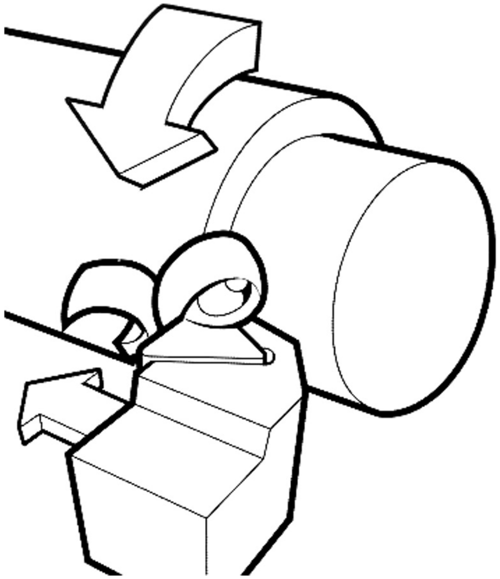
LEARNING TARGETS

The student knows:

- basic principles, tools and machine tools;
- how to evaluate cutting force, power and work;
- how to define the appropriate process parameters.

TURNING

Single point cutting tool removes material from a rotating workpiece to generate a rotational part.



The workpiece owns the cutting motion (rotational).

The tool owns the feed motion.

Types of Turning Operations

Facing

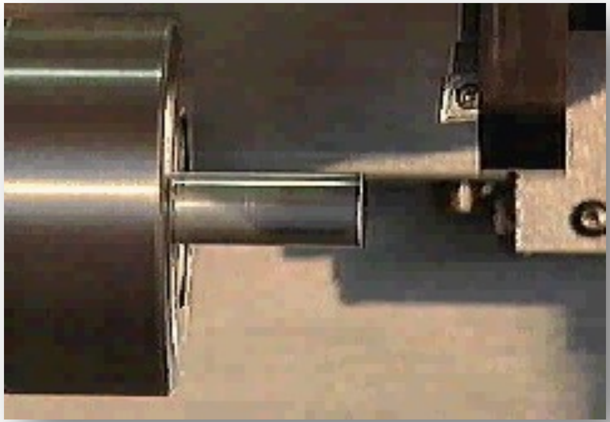
Profiling

Drilling

Types of Turning Operations

Boring

Profiling

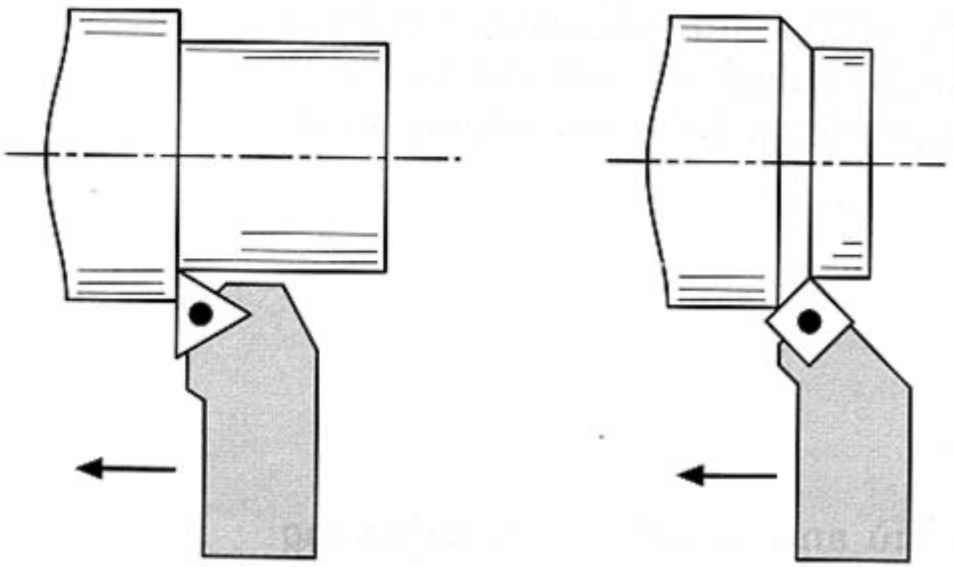


Profiling

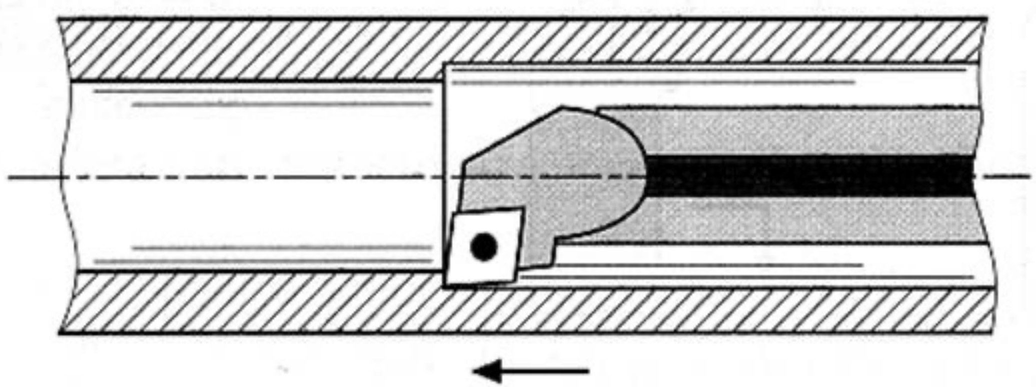
Cutting-off

Types of Turning Operations

External cylindrical turning (straight turning)

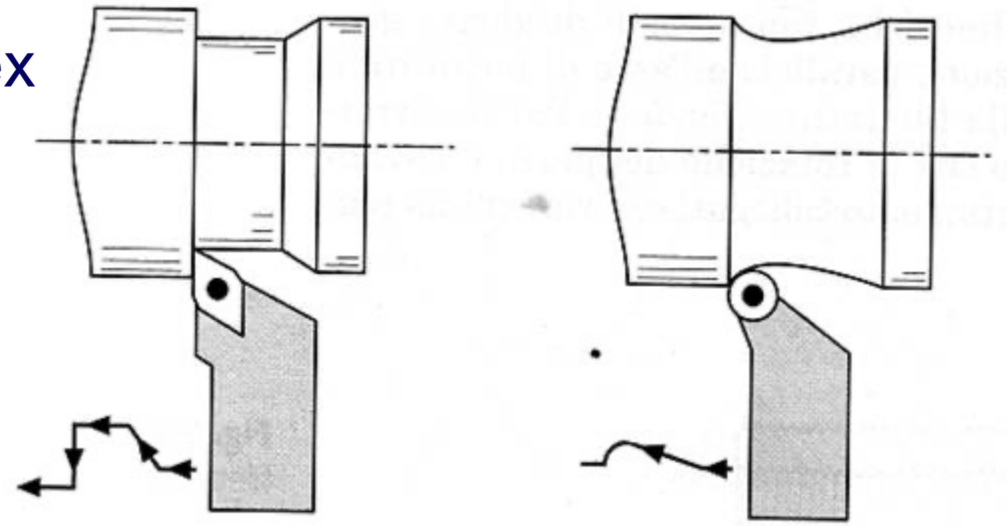


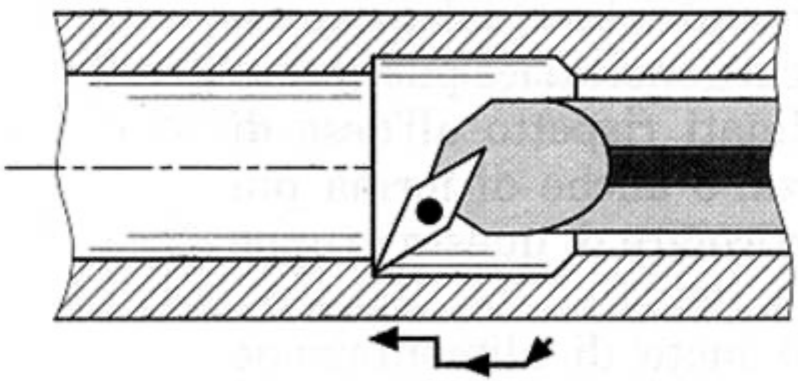
Internal cylindrical turning



TYPES OF TURNING OPERATIONS

External turning of complex surfaces (profiling or contour turning)

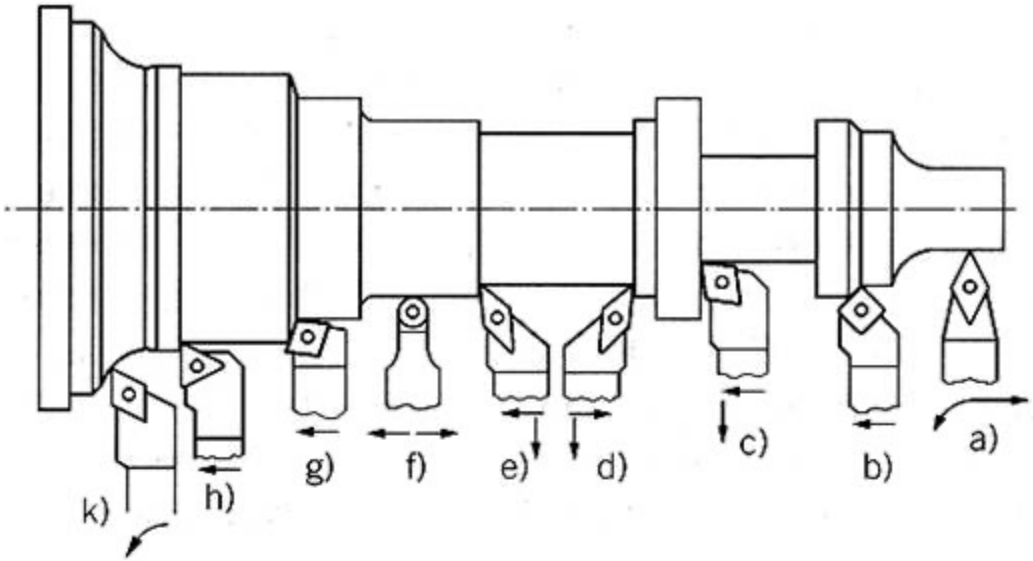




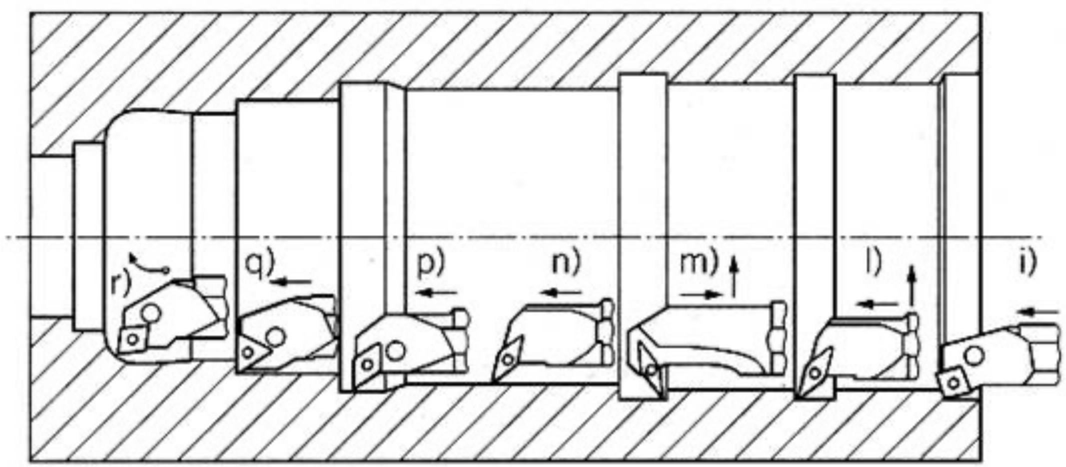
Internal turning of complex surfaces

TYPES OF TURNING OPERATIONS

External turning

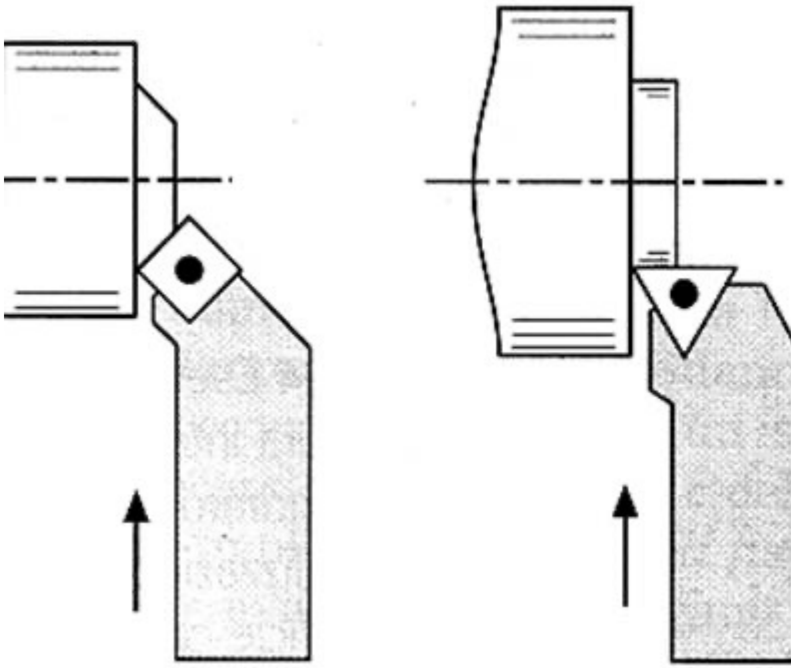


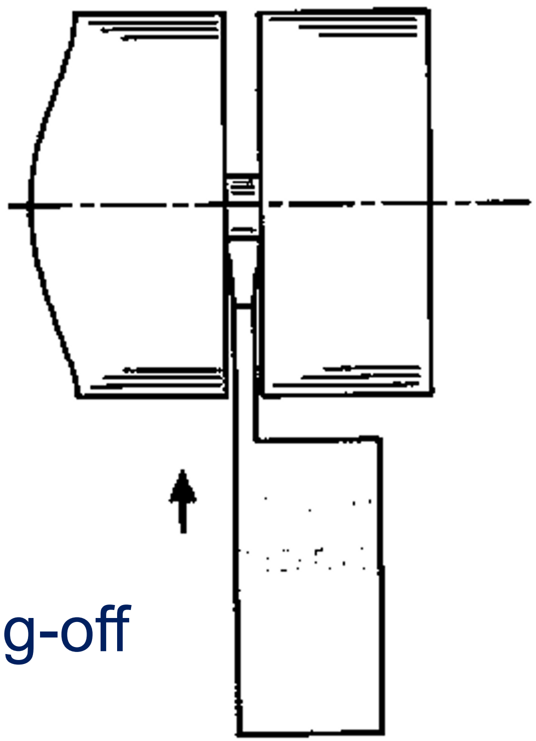
Internal turning



TYPES OF TURNING OPERATIONS

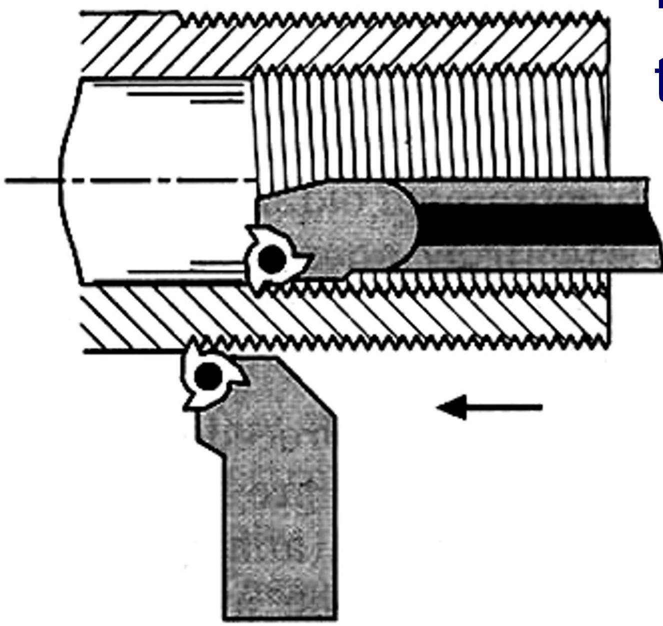
Facing





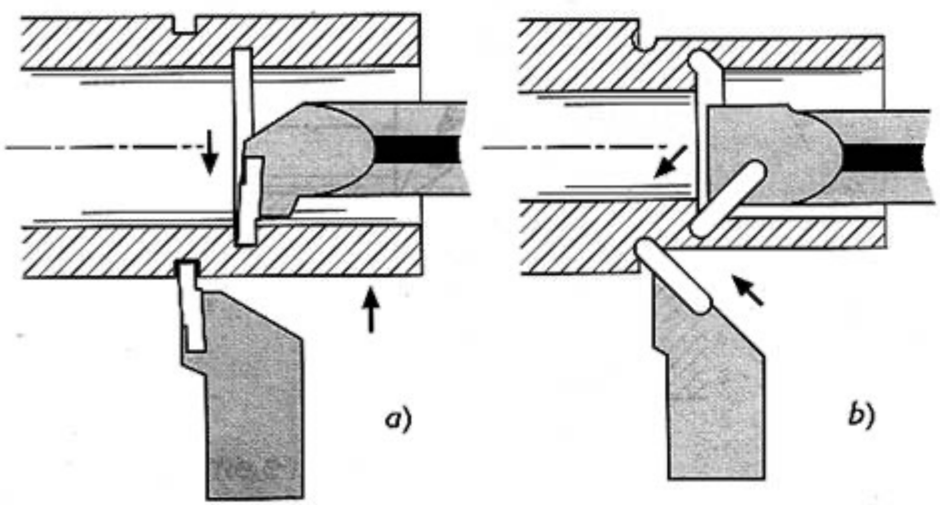
Cutting-off

TYPES OF TURNING OPERATIONS

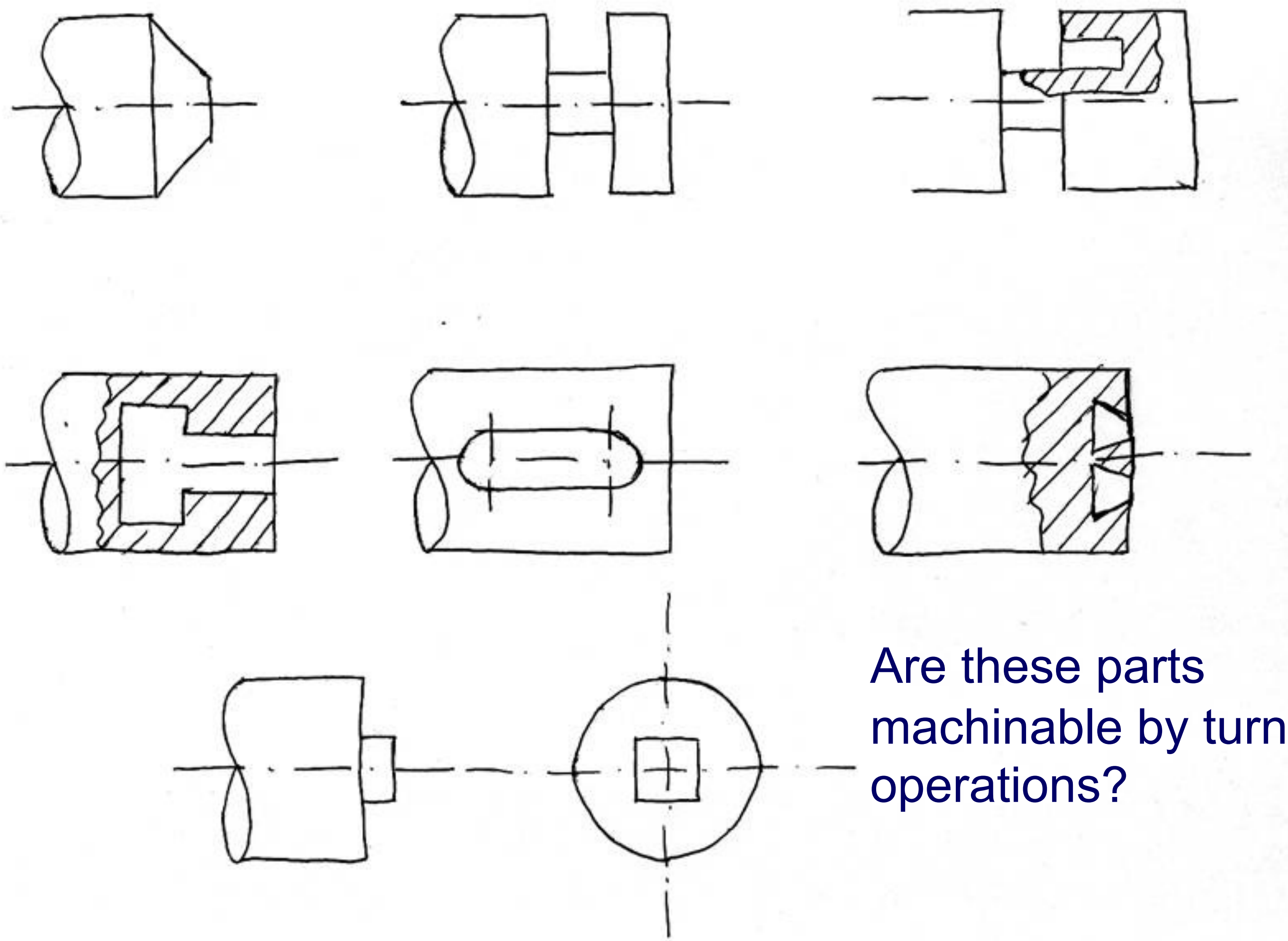


External and internal threading

Grooving



TYPES OF TURNING OPERATIONS



Are these parts machinable by turning operations?

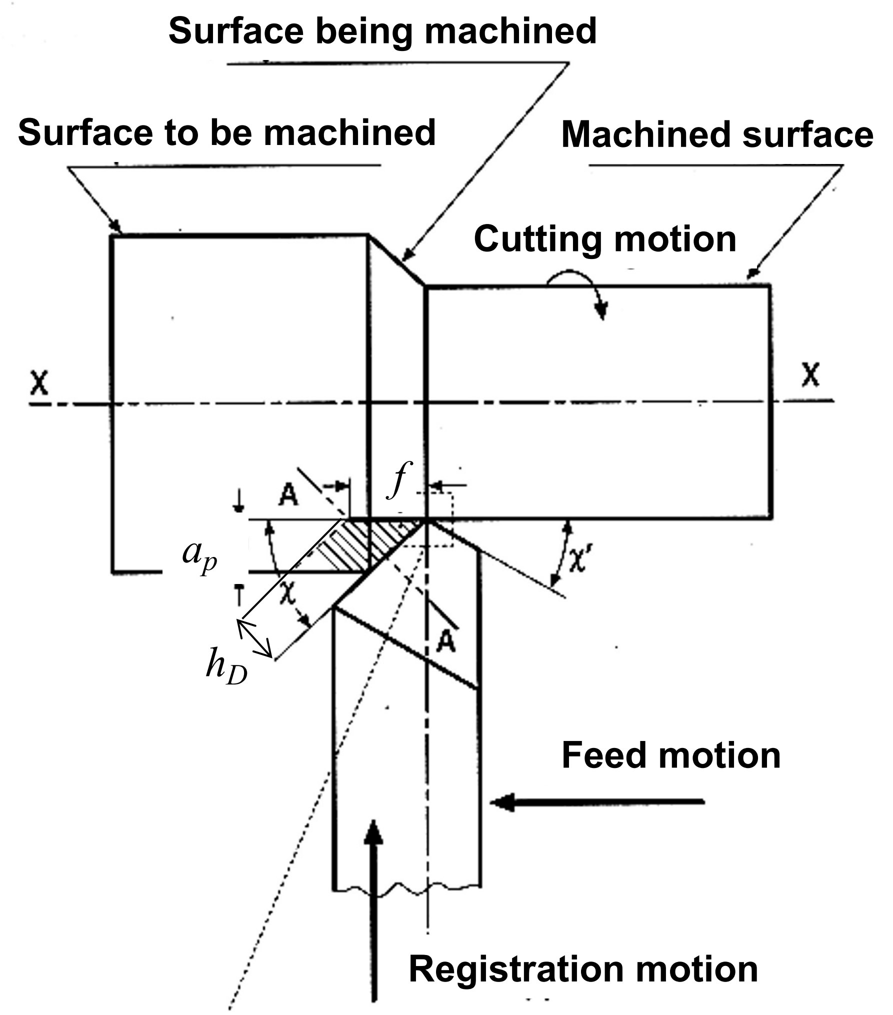
Turning Parameters

- Machined diameter: D [mm]
- Feed per revolution: f [mm/rev]
- Depth of cut: a_p [mm]
- Chip thickness: h_D [mm]
- Chip section: A_D [mm^2]
- Cutting speed: v_c [m/min]
- Rotational speed: n [rpm]

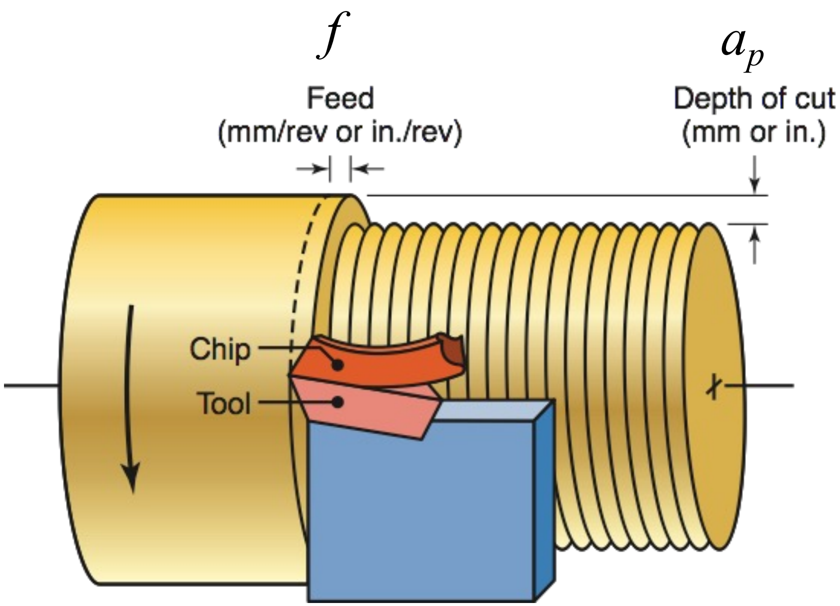
$$v_c = \pi D n / 1000 \quad [\mathrm{m/min}]$$

$$v_f = f n \quad [\mathrm{mm/min}]$$

$$A_D = f a_p \quad [\mathrm{mm}^2]$$



TURNING PARAMETERS

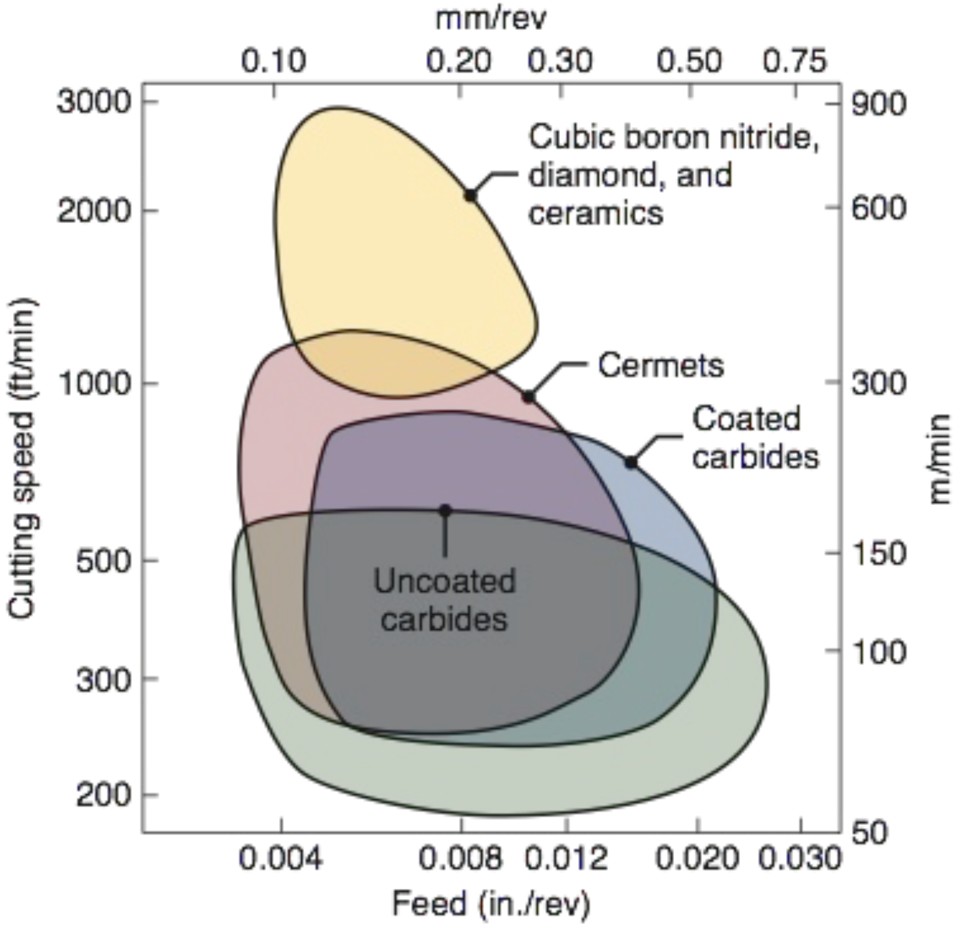


TURNING PARAMETERS

Depth of cut

SANDVIK
Coromant

Cutting Speeds for Turning

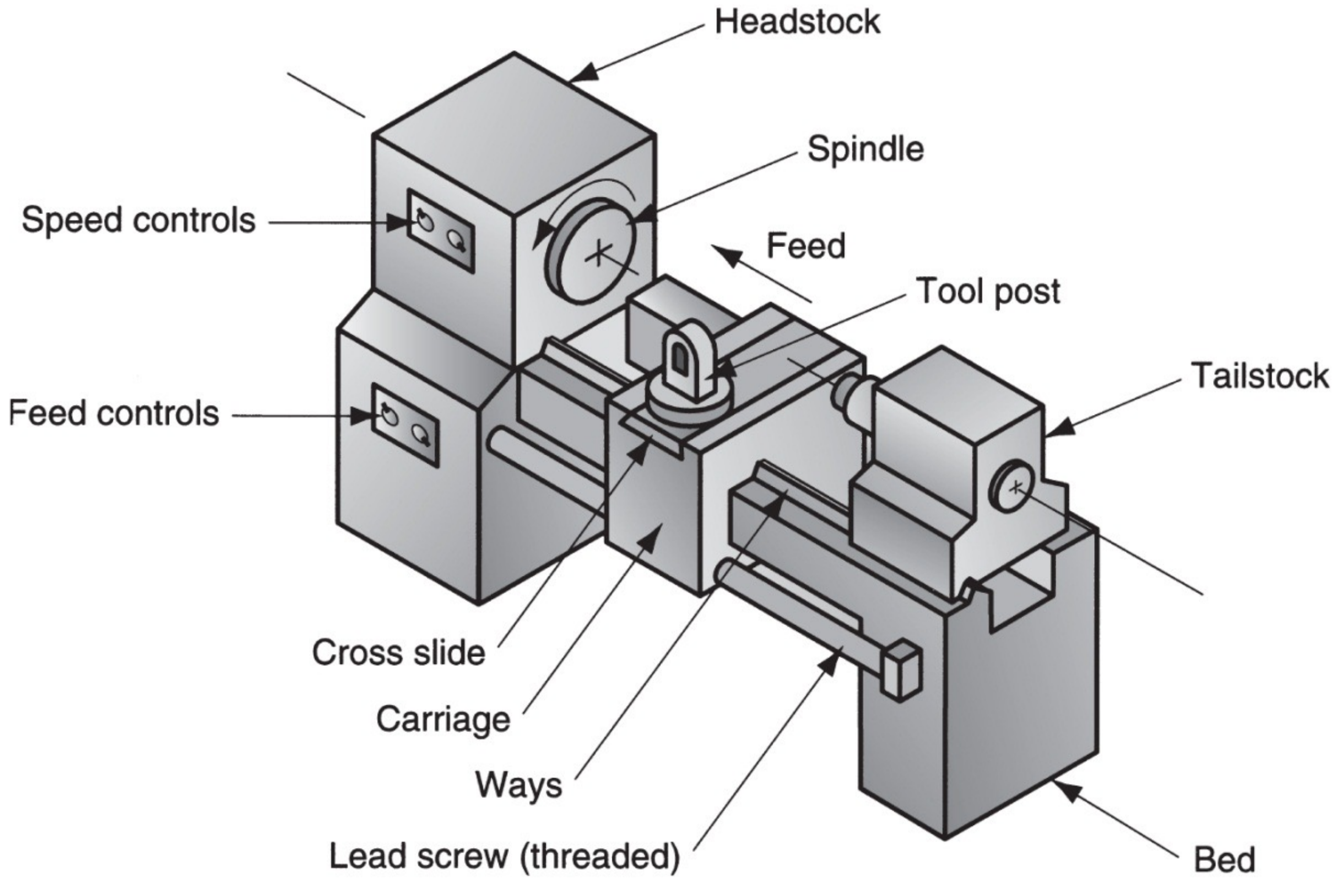


Workpiece Material	Cutting Speed m/min	Cutting Speed ft/min
Aluminum alloys	200-1000	650-3300
Cast iron, gray	60-900	200-3000
Copper alloys	50-700	160-2300
High-temperature alloys	20-400	65-1300
Steels	50-500	160-1600
Stainless steels	50-300	160-1000
Thermoplastics and thermosets	90-240	300-800
Titanium alloys	10-100	30-330
Tungsten alloys	60-150	200-500

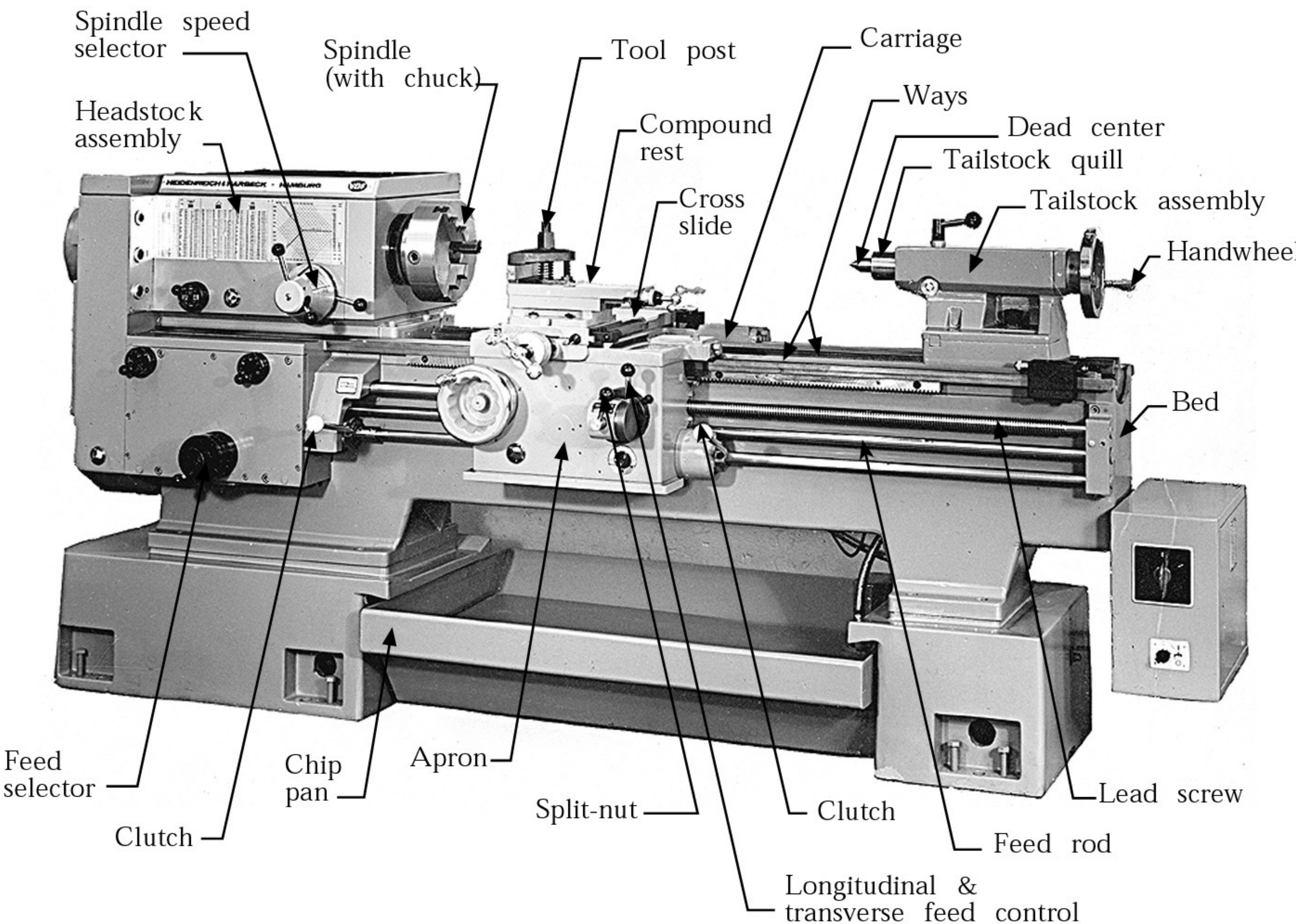
Note: (a) The speeds given in this table are for carbides and ceramic cutting tools. Speeds for high-speed-steel tools are lower than indicated. The higher ranges are for coated carbides and cermets. Speeds for diamond tools are significantly higher than any of the values indicated in the table.
(b) Depths of cut, d, are generally in the range of 0.5-12 mm (0.02-0.5 in.).
(c) Feeds, f, are generally in the range of 0.15-1 mm/rev (0.006-0.040 in./rev).

MACHINE TOOL: THE LATHE

Diagram of a lathe showing its principal components and motions



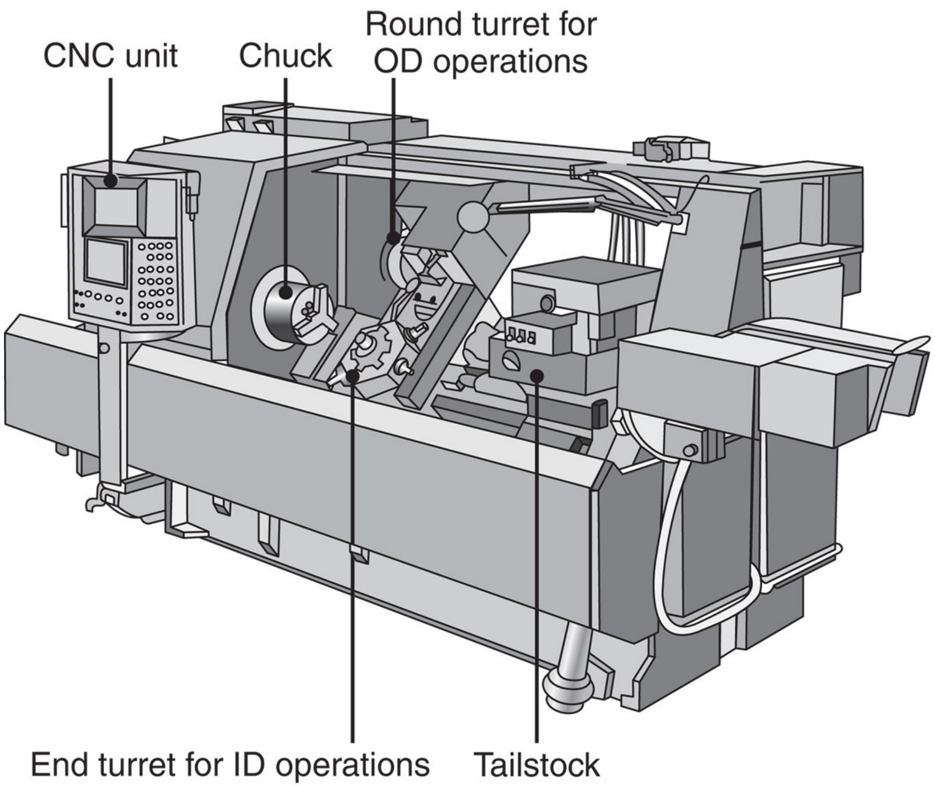
MACHINE TOOL: THE LATHE



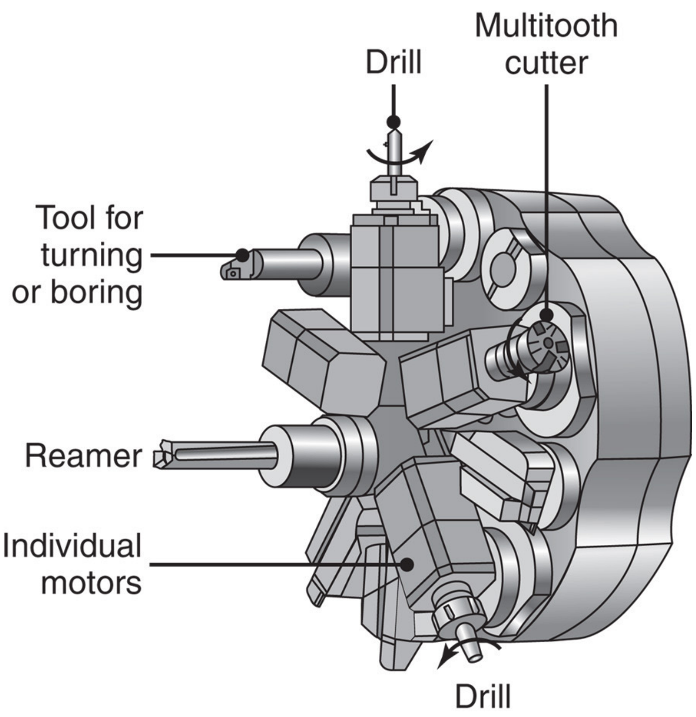
MACHINE TOOL: THE LATHE

(a) A CNC (computer numerical control) lathe.

(b) A typical turret equipped with 10 tools, some of which are powered.



(a)

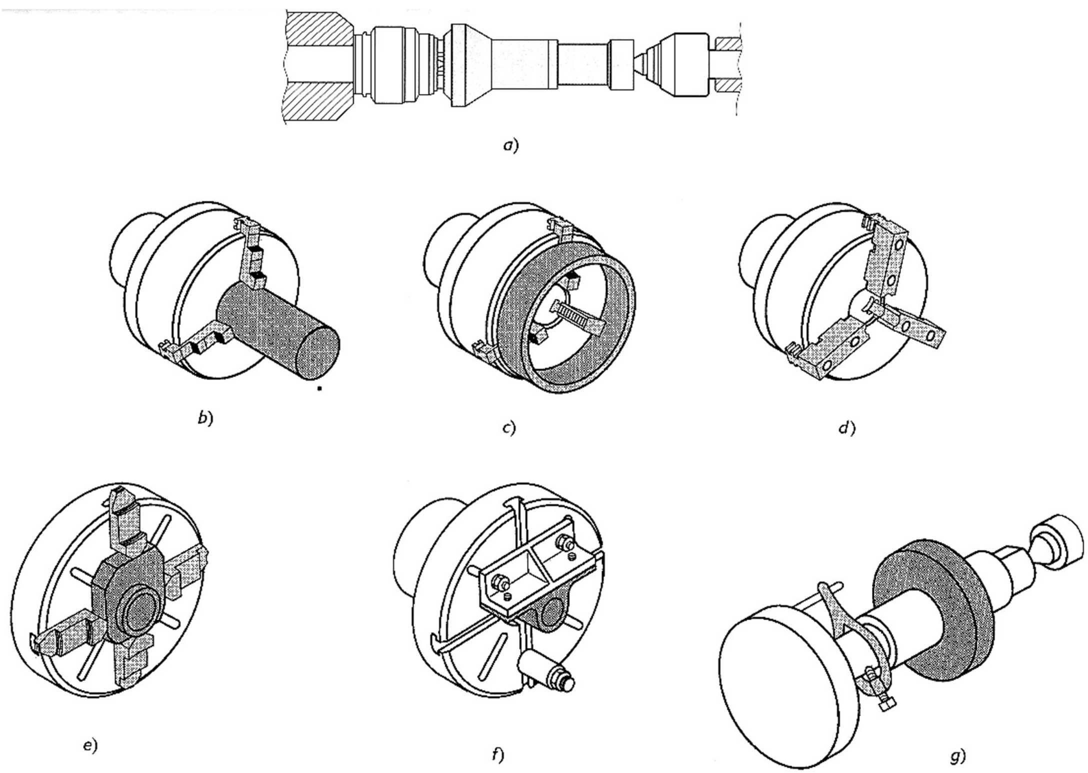


(b)

WORK-HOLDING DEVICES

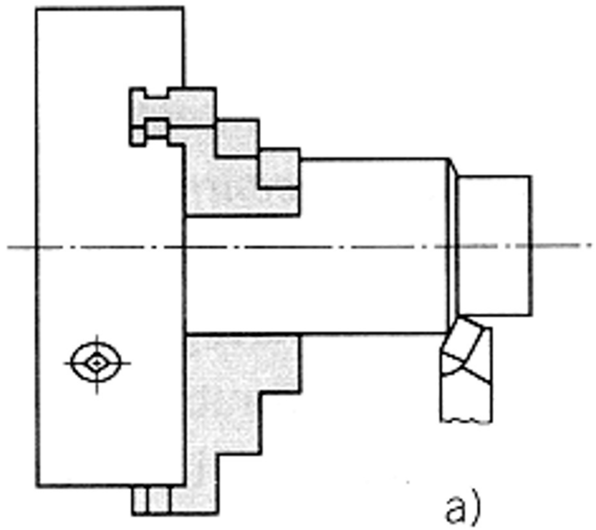
Fixtures are used to:

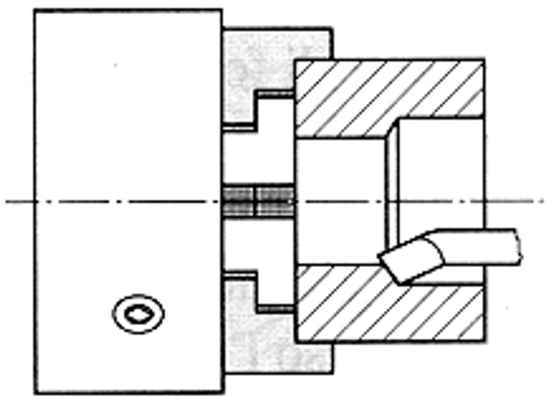
- Support workpiece during the cutting phase;
- Define the position of the part to be machined (*reference*);
- Block the part (*blocking*).



WORK-HOLDING DEVICES

Self-centering three-jaws chuck


a)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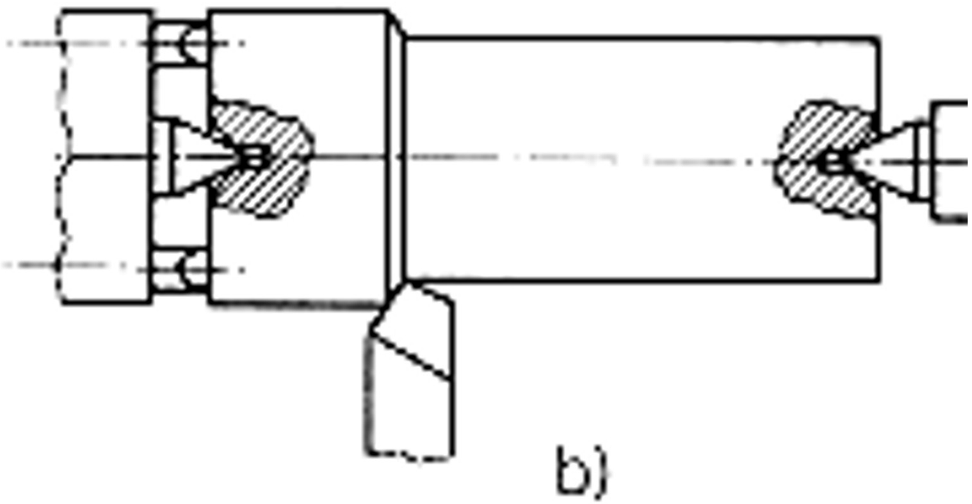
WORK-HOLDING DEVICES

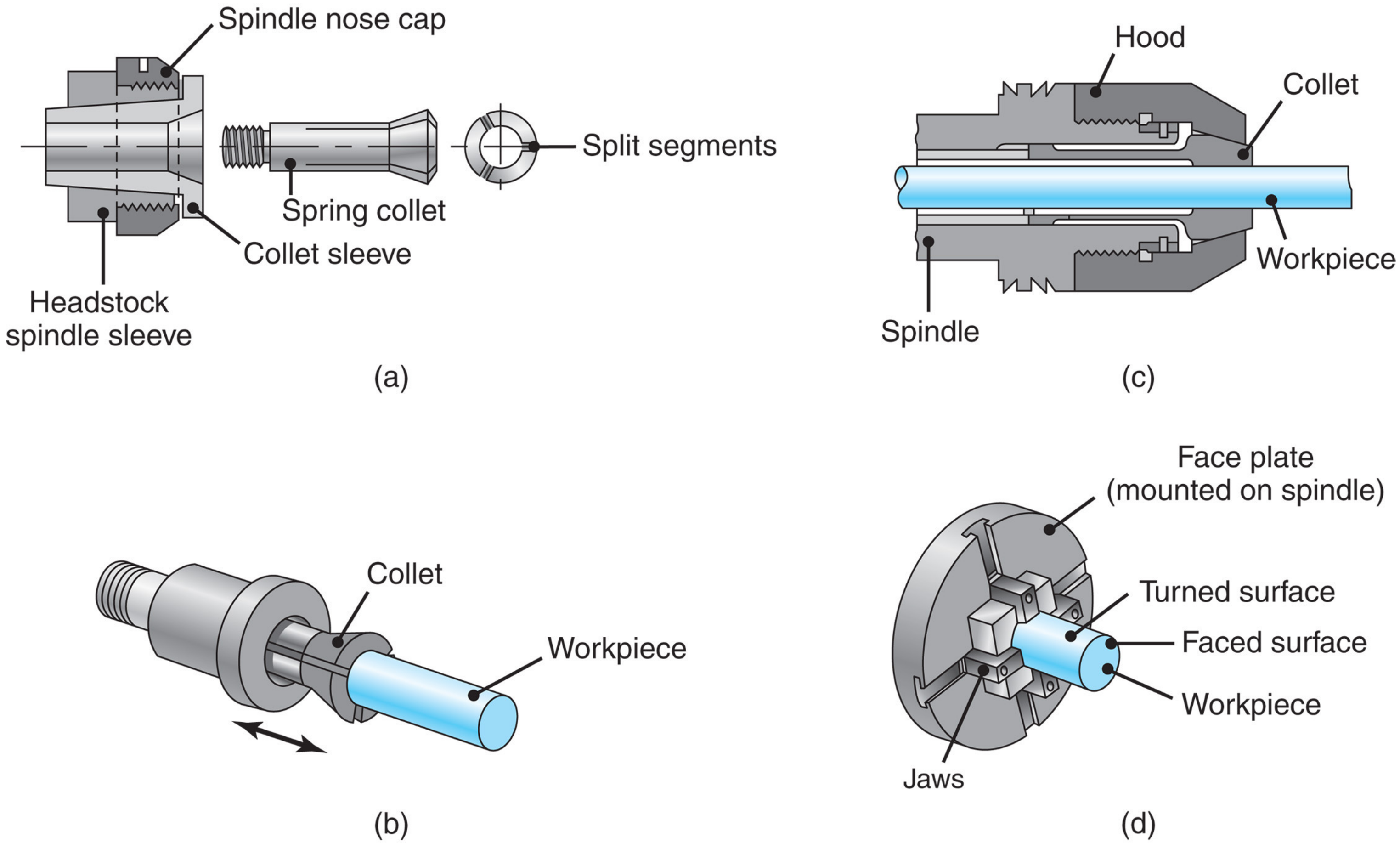
Mounting the work between centers
using frontal teeth



Live center – Dead center

WORK-HOLDING DEVICES

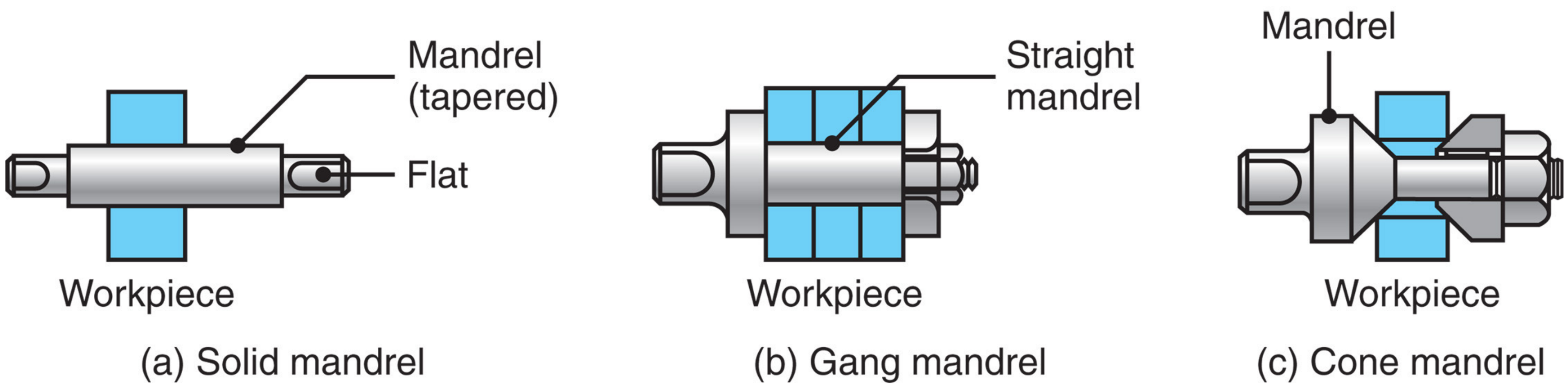
(a) and (b) Schematic illustrations of a draw-in type of ***collet***. The workpiece is placed in the collet hole, and the conical surfaces of the collet are forced inward by pulling it with a draw bar into the sleeve. (c) A push-out type of collet. (d) Work holding of a workpiece on a face plate.



Work-holding Devices

Various types of ***mandrels*** to hold workpieces for turning, these mandrels usually are mounted between centers on a lathe.

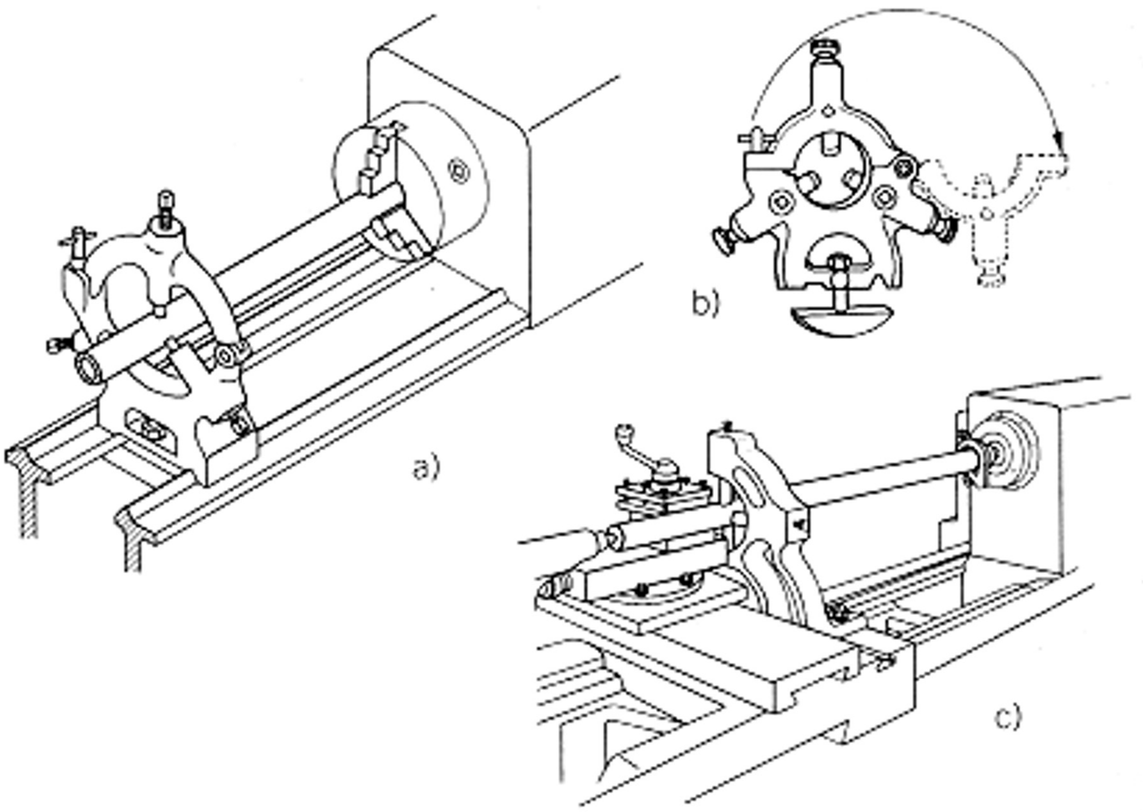
Note that in (a) both the cylindrical and the end faces of the workpiece can be machined, whereas in (b) and (c) only the cylindrical surfaces can be machined.



Work-holding Devices

Rest is a work-holding device, which is used to hold very long workpieces, so to prevent the deflection.

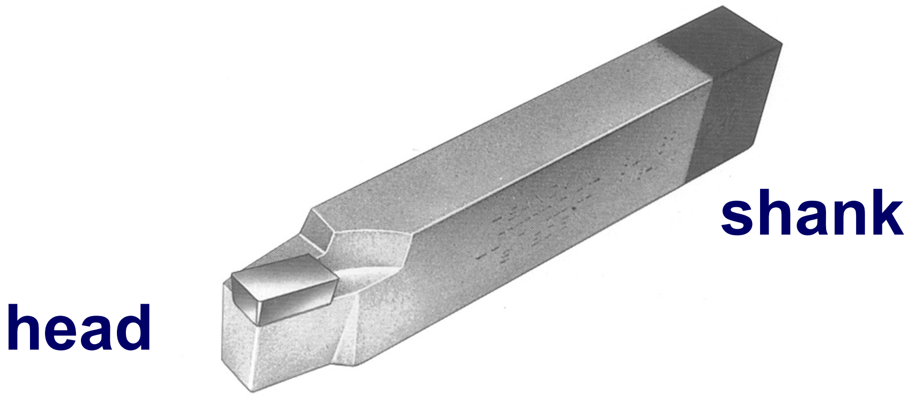
Rests are classified as *steady rest* or *follower rest*.



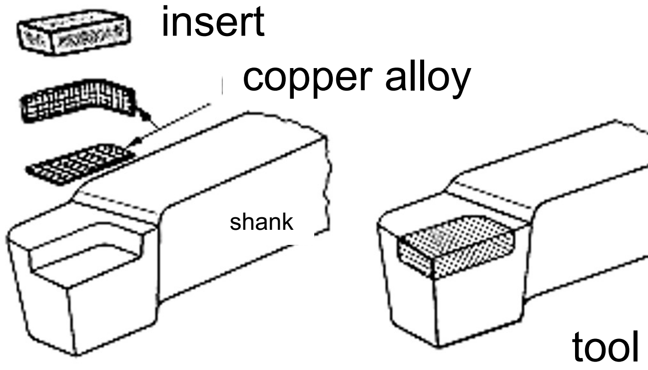
THE TOOL

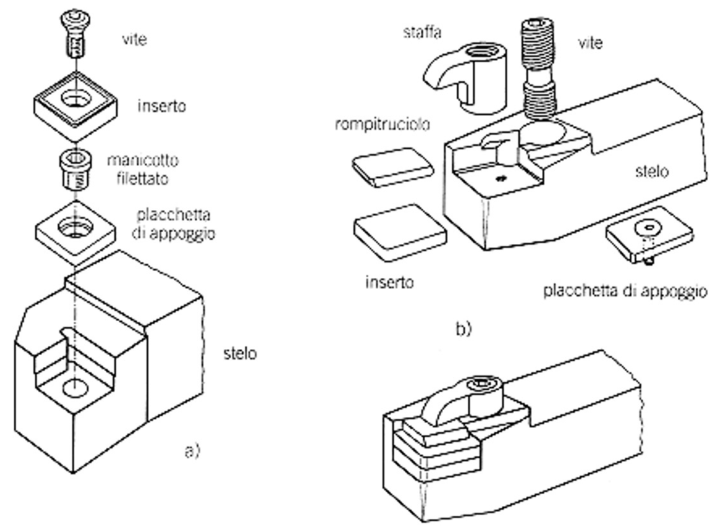
A simple single-point cutting tool has:

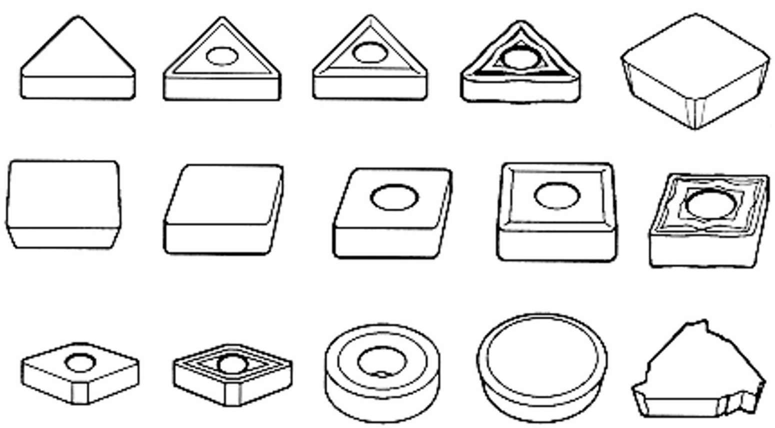
- A **shank**, that allows to fix it on the machine tool;
- A **cutting head**, the portion that gets in touch with the workpiece to be machined:
  - It can be of the same material of the shank,
  - It can be of a different material brazed on the shank,
  - It can be of a different material mechanically fixed on the shank (inserts).



THE TOOL





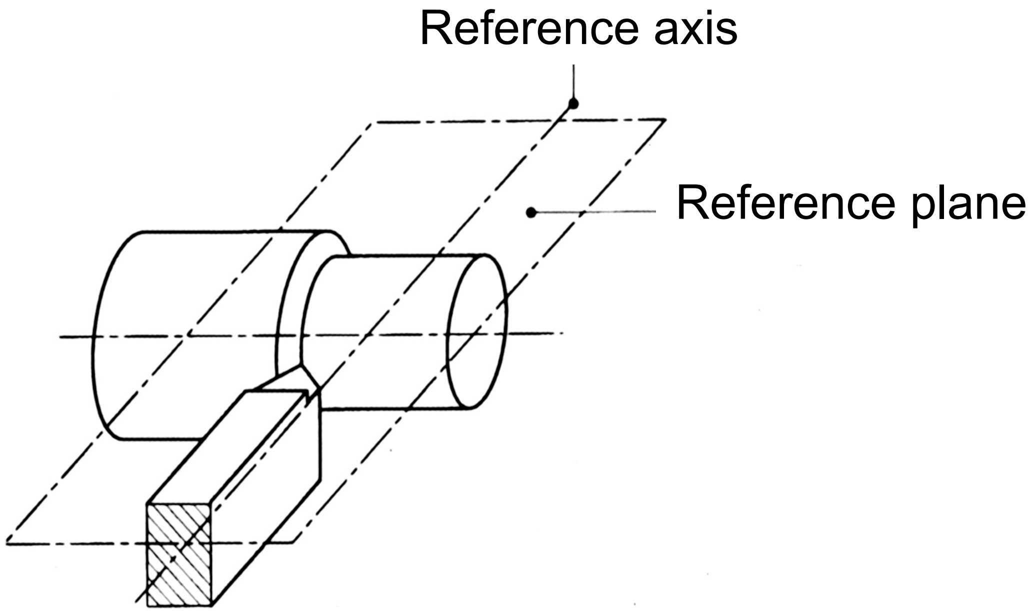


Types of inserts

TOOL GEOMETRY

Reference system

- Shank axis
- Plane through the tool tip and parallel to the shank base



Main Characteristic Angles

Normal section angles:

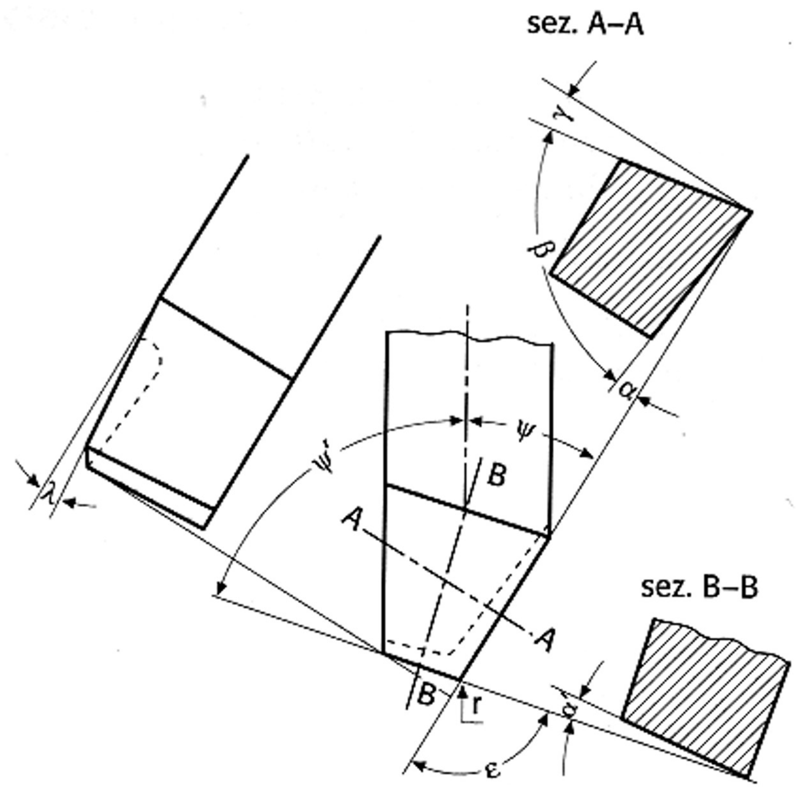
- γ_0 rake angle
- α_0 clearance or relief angle
- α' secondary clearance angle
- β_0 solid angle

Profile angles:

- Ψ main cutting edge angle
- Ψ' secondary cutting edge angle
- ε cutting edges' angle
- λ back rake angle

Entering angles:

- κ main cutting edge entering angle
- κ' secondary cutting edge entering angle

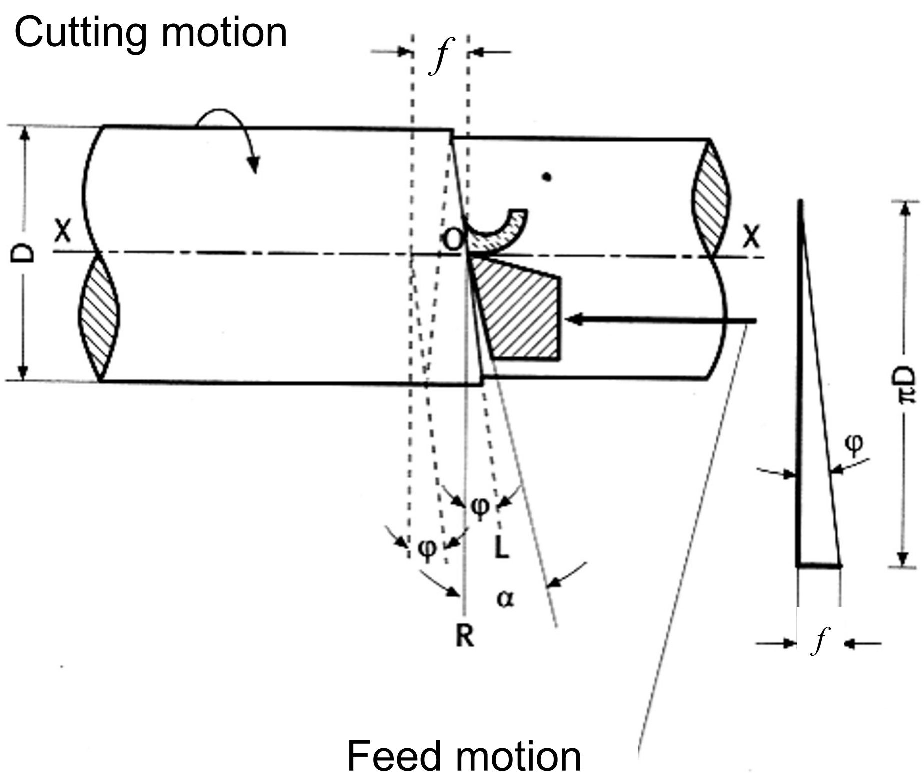


Clearance or Relief Angle

During turning the tool describes a helix of pitch f and diameter *D*.

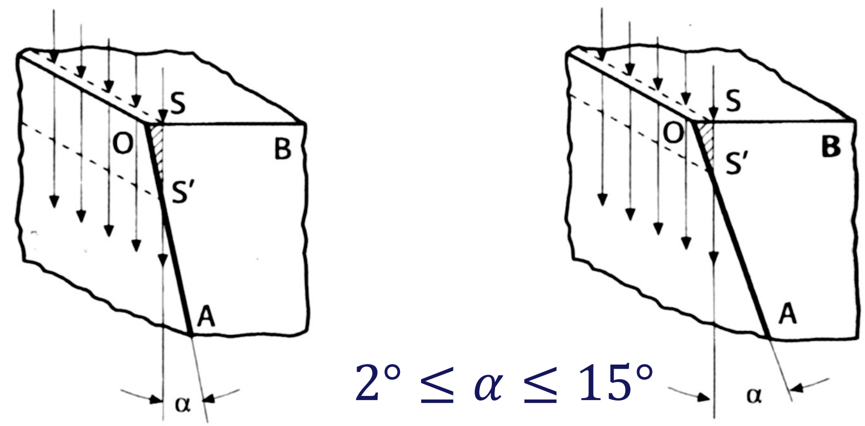
The OL trace is inclined by an angle φ that reduces the size of α_0.

$\alpha_0 - \varphi$ must be positive, to avoid contact and friction between the tool flank and the machined surface.

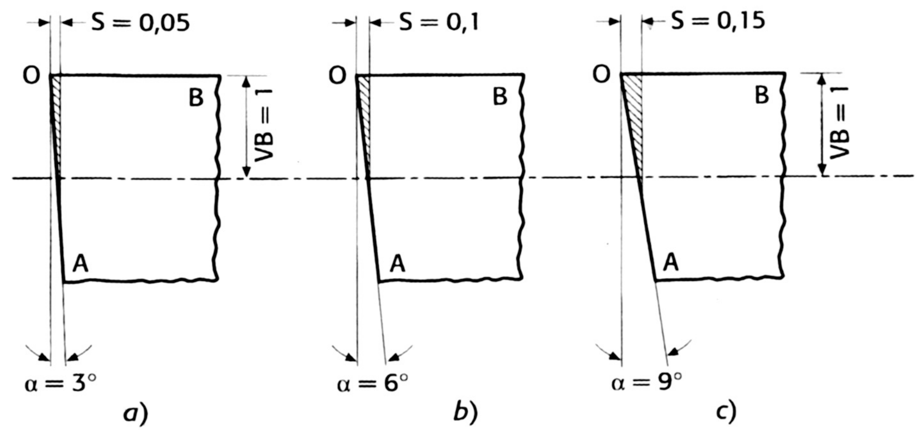


CLEARANCE OR RELIEF ANGLE

- If α is too high the resistant section is too little.
- Too low α_0 causes a faster reaching of the maximum allowed flank wear.
- It depends on the material to machine (the higher the cutting pressure the lower should α_0 be), and on the material of the tool (if not much tough, the values of α_0 should be lower).



$$2° \leq \alpha \leq 15°$$

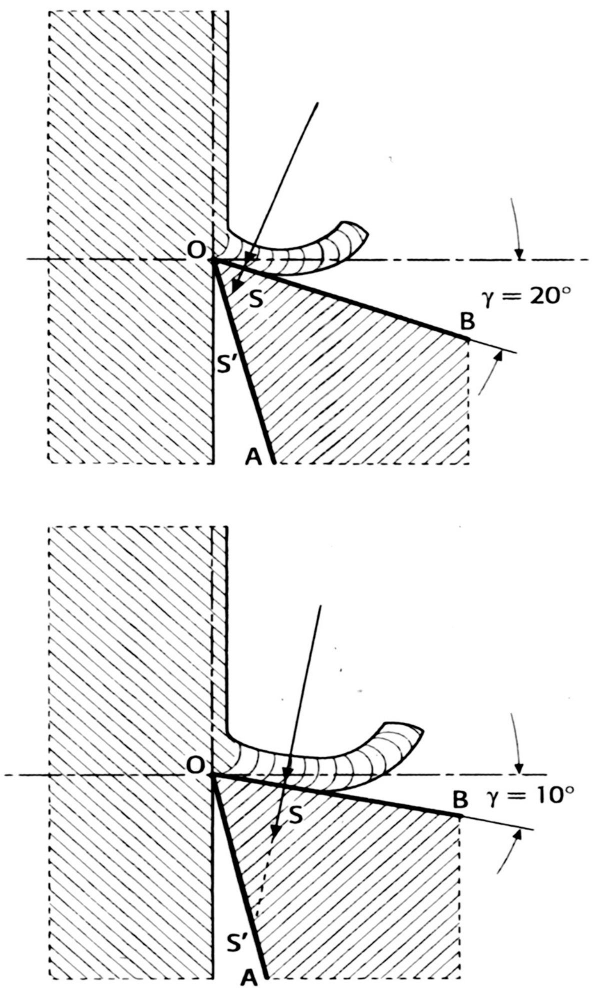


RAKE ANGLE

γ_0 affects the chip formation mechanism.

Remaining unchanged the other conditions, higher γ_0 means sharper cutting edge and determines lower:

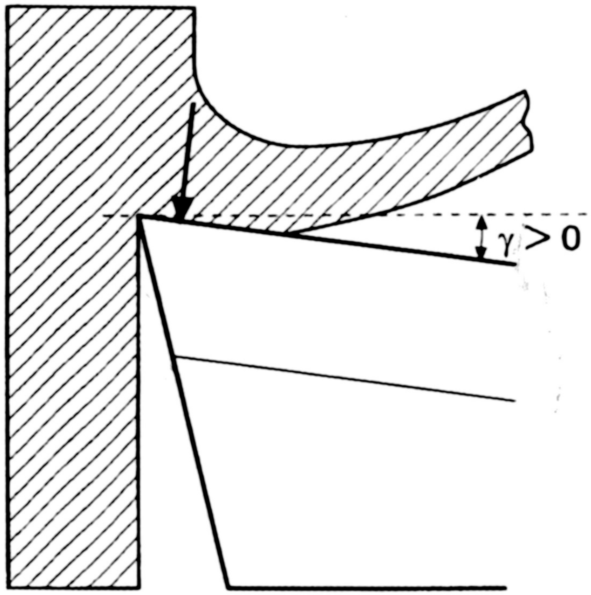
- Resistance of the cutting edge,
- Chip deformations,
- Cutting pressure,
- Sliding friction on the rake,
- Forces,
- Power and temperatures.

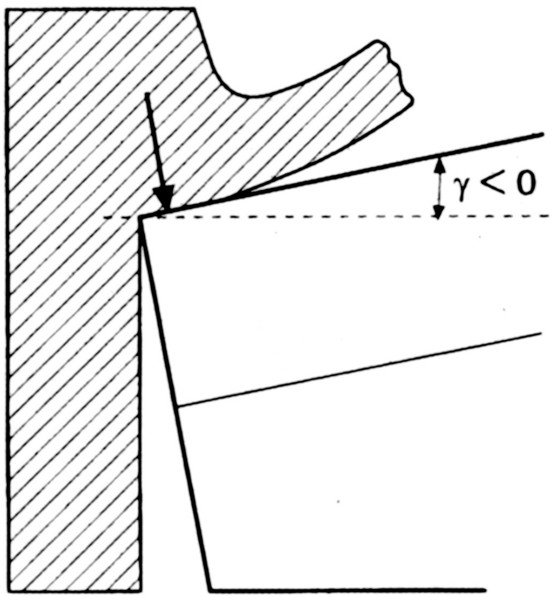


Rake Angle

- Ductile workpiece materials require higher rake angles.
- Brittle workpiece materials require high resistant sections, therefore lower and even negative rake angles.

γ_0 negative determines increase of forces, temperatures and absorbed power.





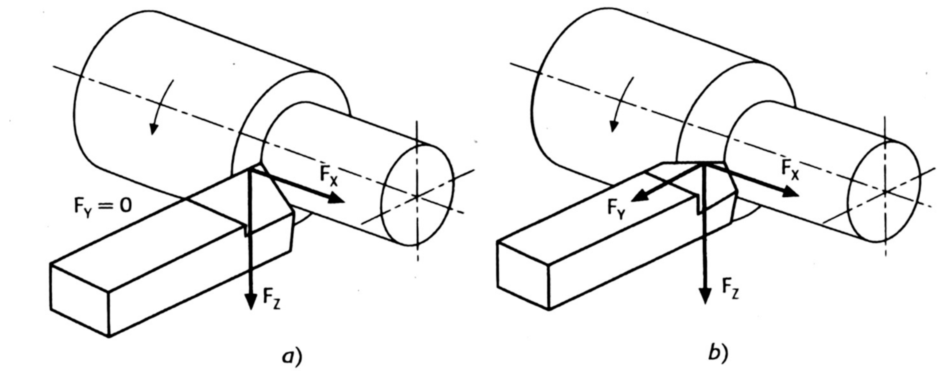
MAIN CUTTING EDGE ANGLE

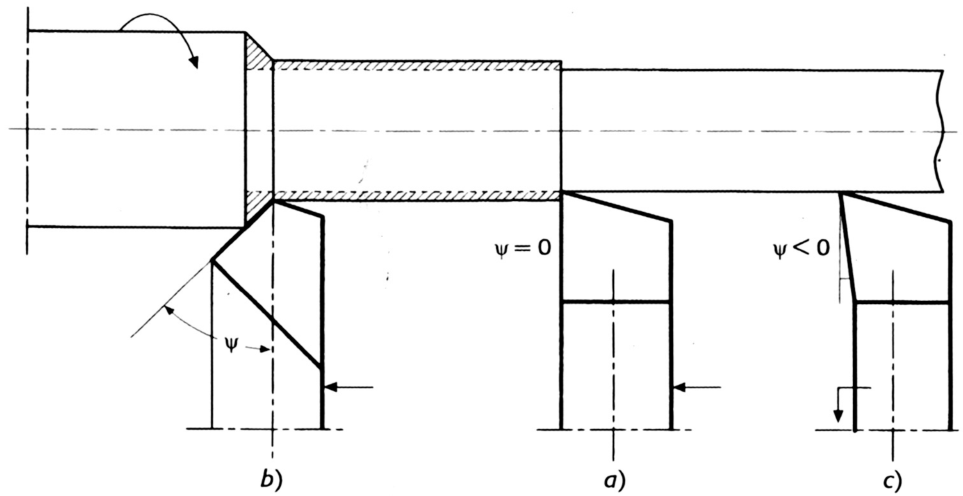
Ψ affects the components of the cutting force.

High Ψ allows for higher cutting edge duration, but determines an increase of the thrust force (repulsive force).

Usually:

- Ψ = 0 or < 0 with flexible workpiece or to cut shoulders,
- Ψ > 0 in normal working conditions and with rigid machine tool-workpiece systems.





PROFILE ANGLES

Ψ' secondary cutting edge angle

- Together with Ψ determines the angle ε of the cutting edges and, therefore, the robustness of the tool nose.
- Ψ' should have high values, compatibly with the secondary cutting edge registration angle, which affects the surface roughness.

ε cutting edge's angle (tool nose angle)

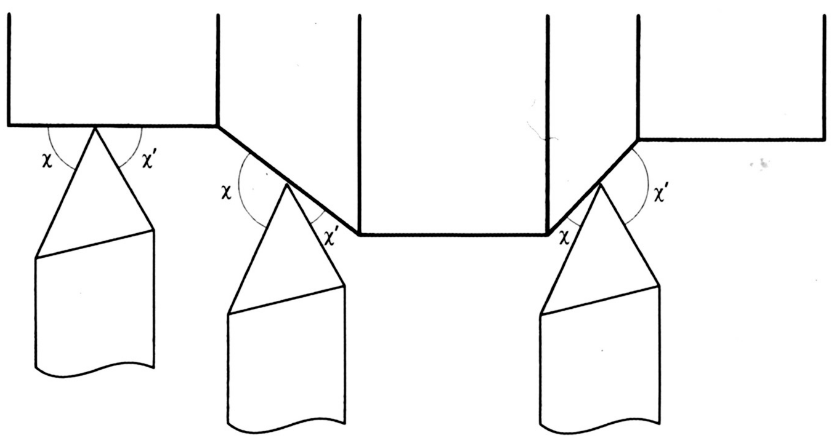
- It affects the tool nose robustness and the surface finish of the part.

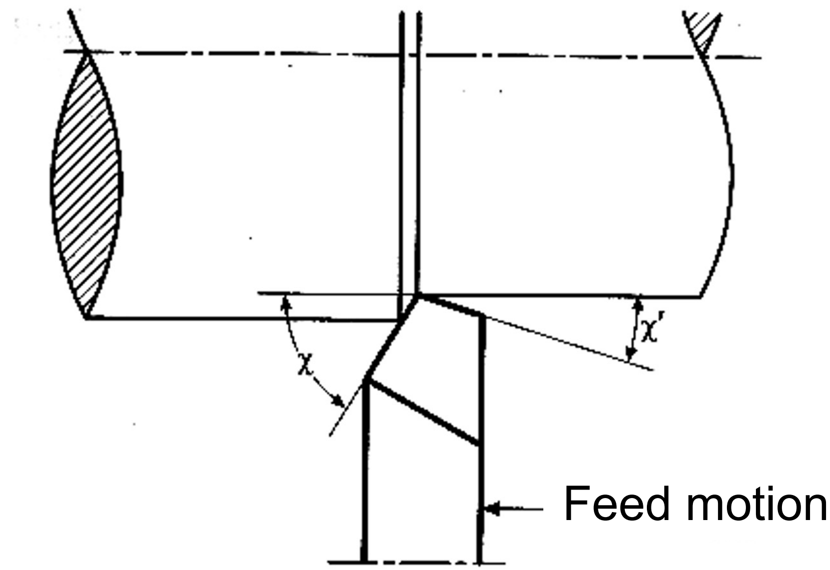
ENTERING ANGLES

κ Entering angle of the main cutting edge

κ' Entering angle of the secondary cutting edge

Together with feed and nose radius of the tool they determine the theoretical roughness of the machined surface.

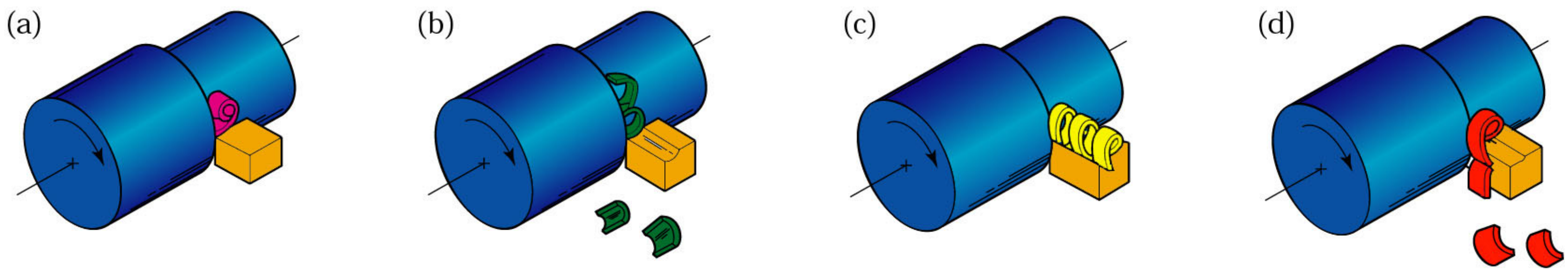




RECOMMENDATIONS

Material	High-speed steel					Carbide inserts				
	Back rake	Side rake	End relief	Side relief	Side and end cutting edge	Back rake	Side rake	End relief	Side relief	Side and end cutting edge
Aluminum and magnesium alloys	20	15	12	10	5	0	5	5	5	15
Copper alloys	5	10	8	8	5	0	5	5	5	15
Steels	10	12	5	5	15	-5	-5	5	5	15
Stainless steels	5	8-10	5	5	15	-5-0	-5-5	5	5	15
High-temperature alloys	0	10	5	5	15	5	0	5	5	45
Refractory alloys	0	20	5	5	5	0	0	5	5	15
Titanium alloys	0	5	5	5	15	-5	-5	5	5	5
Cast irons	5	10	5	5	15	-5	-5	5	5	15
Thermoplastics	0	0	20-30	15-20	10	0	0	20-30	15-20	10
Thermosets	0	0	20-30	15-20	10	0	15	5	5	15

Angles and Chip Formation



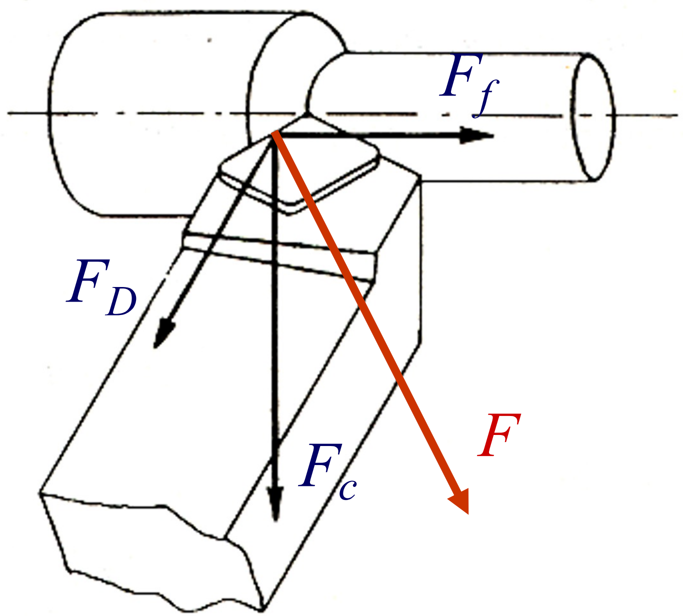
(a) tightly **curled** chip

(b) chip **hits** workpiece and breaks

(c) **continuous** chip moving away from workpiece

(d) chip **hits** tool shank and breaks off

FORCES



F_f = Feed force

F_D = Thrust force

F_c = Cutting force

CUTTING POWER

We compute the scalar product:

$$P = \vec{F} \times \vec{v} = F_c v_c + F_f v_f + F_D v_D$$

but, since $v_D = 0$, it follows that:

$$P = F_c v_c + F_f v_f$$

Being

$$v_f << v_c \qquad F_f < F_c$$

it follows that:

$$P = P_c = F_c v_c$$

Cutting Force

In order to predict the cutting force F_c we use the cutting pressure k_c method:

$$F_c \overset{\text{def}}{=} k_c A_D$$

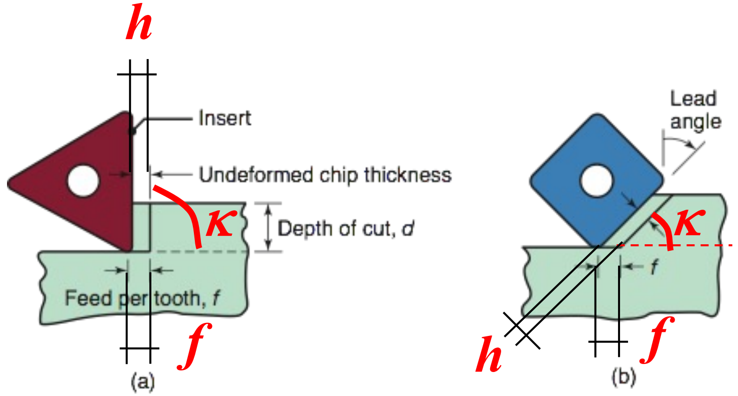
Where:

$$k_c = \frac{k_{cs}}{h_D^x}$$

$$A_D = f\, a_p = h_D\, b$$

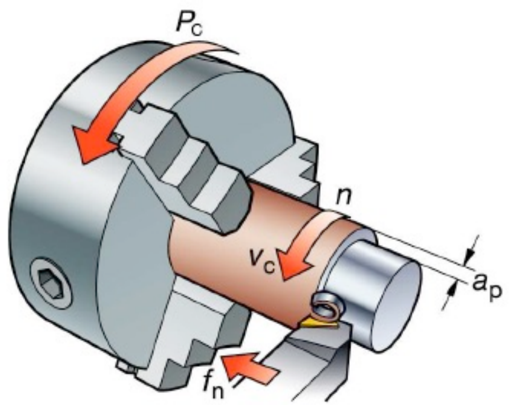
Cutting Force

It is more convenient to express F_c as a function of f and a_p:



$$h_D = f \sin\kappa$$

$$k_c = \frac{k_{cs}}{h_D^x} = \frac{k_{cs}}{f^x(\sin^x\kappa)} = \frac{k_{cs}}{f^x}\left(\frac{1}{\sin\kappa}\right)^x$$



$$F_c = k_c A_D = k_c f a_p = k_{cs} f^{1-x} a_p \left(\frac{1}{\sin\kappa}\right)^x$$

Cutting Force

On tool manufacturers' catalogues you may find different values, e.g. $k_{c\,0,4}$ (cutting pressure for f = 0,4 mm/rev and κ = 90°), and not k_{cs}:

$$k_c = \frac{k_{cs}}{h_D^x} = \frac{k_{cs}}{f^x(\sin\kappa)^x} = \frac{k_{cs}}{0,4^x(\sin 90°)^x} = \frac{k_{cs}}{0.4^x} = k_{c0,4}$$

$$\Rightarrow \quad k_{cs} = k_{c\,0,4}\, 0,4^x$$

And it follows that:

$$F_c = k_{c\,0,4}\; a_p f^{1-x}\left(\frac{0,4}{\sin\kappa}\right)^x$$

Cutting and Machining Time

Considering a length of cut equal to l_c, we define:

$$t_c = \frac{l_c}{v_f} = \frac{l_c}{f \cdot n}$$

Cutting time

$$l_{tot} = l_c + e_{in} + e_{out}$$

Total length of the tool path in working feed

$$t_m = \frac{l_{tot}}{v_f} = \frac{l_{tot}}{f \cdot n}$$

Machining time

VERIFICATIONS IN TURNING

To carry out a turning process, it is necessary to verify that:

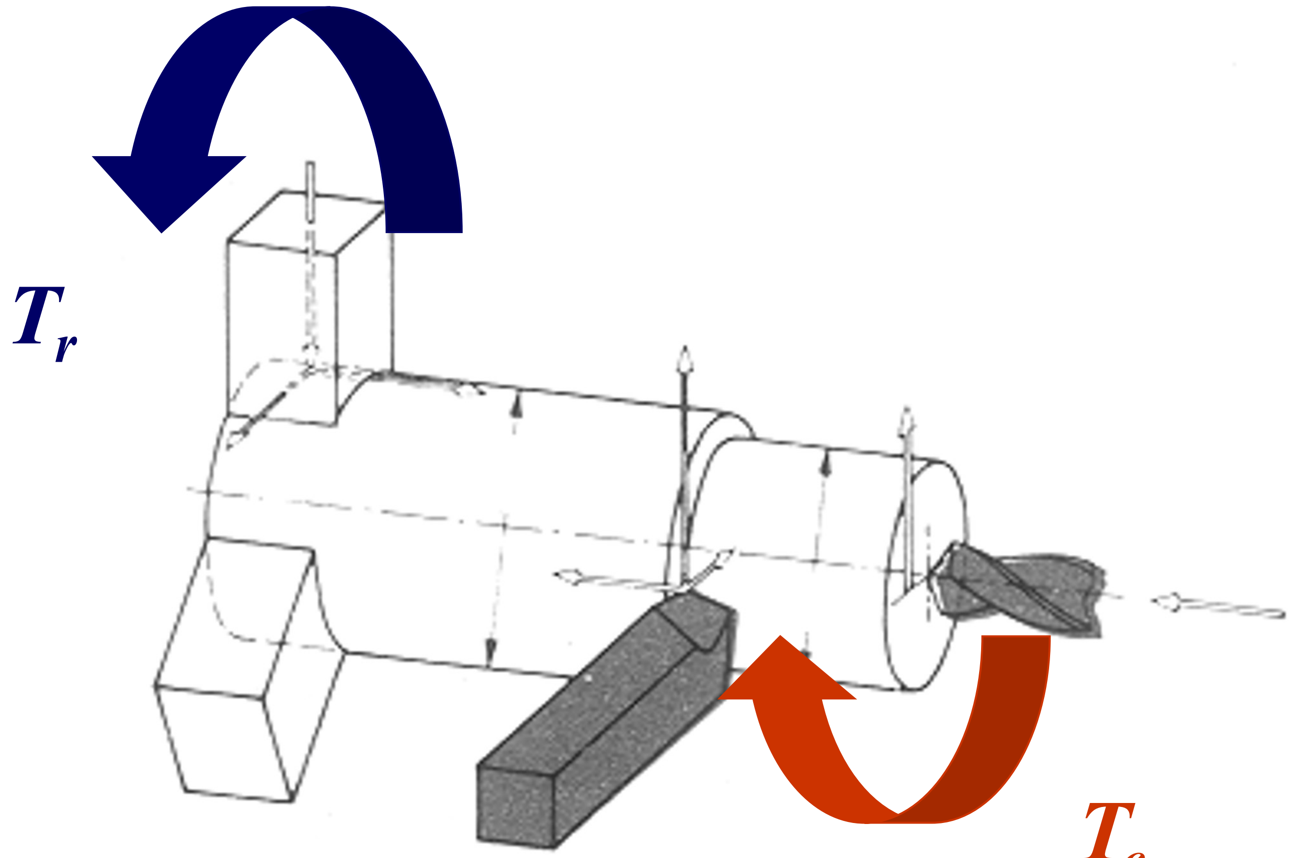
- Cutting depth is compatible with the selected tool;
- The selected feed is admissible for the lathe;
- The cutting speed and, therefore, the selected rotational speed is compatible with the lathe features;
- The power needed for machining is actually deliverable by the considered lathe.

VERIFICATIONS IN TURNING

…furthermore, it is necessary to verify that:

- The cutting parameters are compatible with the required surface finish;
- The cutting parameters are compatible with the required dimensional and geometrical tolerances;
- The selected fixtures are able to hold firmly the workpiece.

FIXTURE VERIFICATION



CHUCK VERIFICATION

Cutting torque: $T_c = F_c D / 2$

Resistant torque: $T_r = z \mu p A D^* / 2$

where:

- z = number of jaws of the chuck;
- p = chuck-workpiece contact pressure;
- A = jaw-workpiece contact area;
- μ = static friction coefficient;
- D = machined diameter;
- D* = workpiece diameter at fixturing position.

$$\mu = \begin{cases} \text{0,15 for mild steel jaws;} \\ \text{0,25 for jaws with wavy surface;} \\ \text{0,35} \div \text{0,8 for quenched steel jaws with ruled surface.} \end{cases}$$

Chuck Verification

Cutting torque ≤ Resistant torque

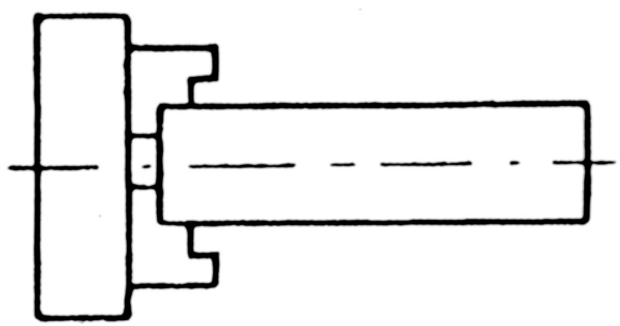
$$T_c = F_c\, D / 2 \;\leq\; T_r = z\, \mu\, p\, A\, D^* / 2$$

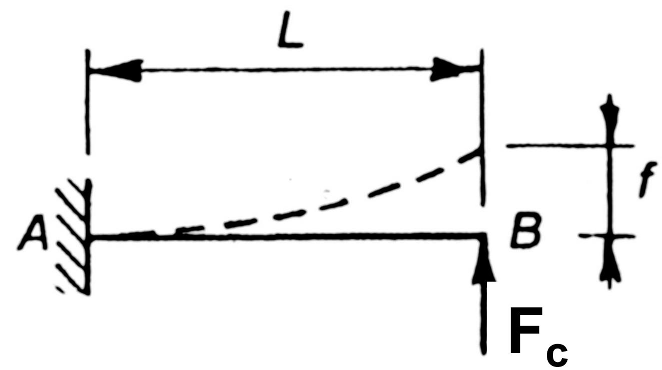
Therefore chuck-workpiece contact pressure should be:

$$p \geq p_{\min} = (F_c\, D) / (z\, \mu\, A\, D^*)$$

Workpiece Bending

Case 1: Three jaws self-centering chuck





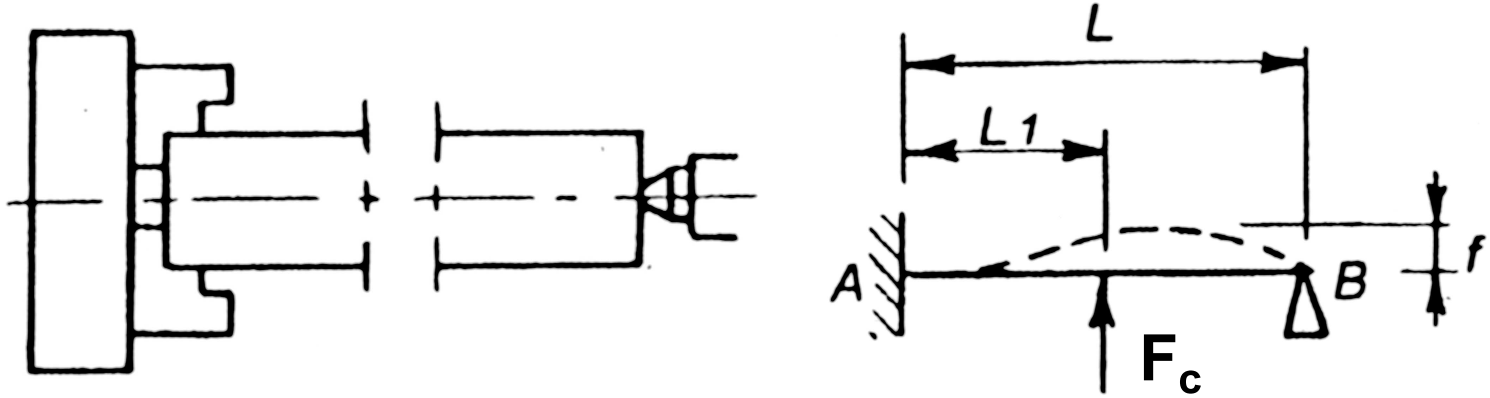
Notice:

The displacement f value is maximum when the force is applied to the free end.

$$f = \frac{1}{3} \cdot \frac{F_c \cdot L^3}{E \cdot J}$$

J: Moment of Inertia of the section

Case 2: Self-centering chuck – Dead center

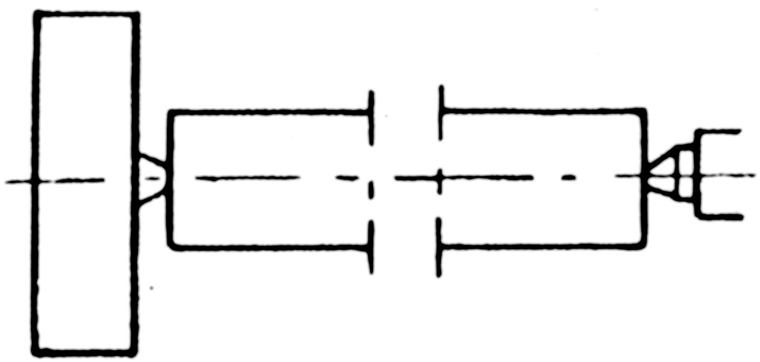


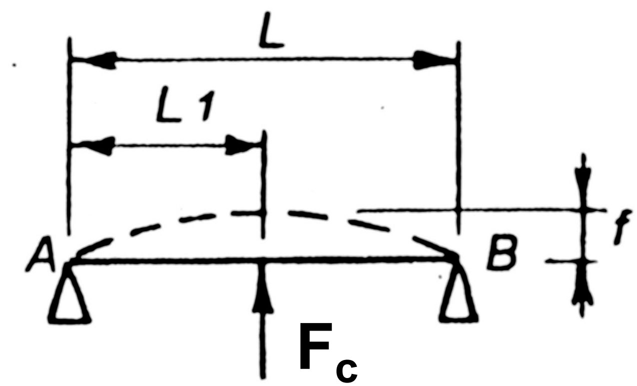
Notice:

The maximum vale of f is obtained when $L_1 \approx 0,6\, L$

$$f = \frac{1}{107} \cdot \frac{F_c \cdot L^3}{E \cdot J}$$

Case 3: Between centers





Notice:

Maximum f is obtained when $L_1 = L/2$

$$f = \frac{1}{48} \cdot \frac{F_c \cdot L^3}{E \cdot J}$$

WORKPIECE BENDING

Case 1: Three-Jaws Chuck

$$f = \frac{1}{3} \cdot \frac{F_c \cdot L^3}{E \cdot J}$$

Case 2: Chuck-Dead center

$$f = \frac{1}{107} \cdot \frac{F_c \cdot L^3}{E \cdot J}$$

Case 3: Between centers

$$f = \frac{1}{48} \cdot \frac{F_c \cdot L^3}{E \cdot J}$$